

Fusion-driven, epigenetically silenced dark matter

in BRAF/RAS-negative thyroid cancer — Paper 1 manuscript v8 navigation hub. 5 Pillars + 11 manuscript files + 8 figures captioned + 17 references + 6-week marathon writing window.

76.8% DM1 FUSION+ ★ RET/NTRK/ALK/BRAF · OR 7.41 vs DM2	2.30 TPO PROMOTER D ★ DM1 epigenetic silencing, fusion-independent	2.53 POOLED OS HR [1.31, 4.89] TCGA + MSK meta, I²=0%	+0.132 ΔAUC VS BRAF 8-gene panel 5-fold CVAUC 0.954
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저자 · SEUNGHO COOK (1ST) CORRESPONDING · YU HYEONG-WON TARGET · CELL REP MED 1순위 → JCI
INSIGHT + NAT COMMUN REACH DATE · 2026-05-02

§ 1 · 5-PILLAR STATUS

5 pillars all STRONG

전체 pillar STRONG 격상 – Cell Rep Med reach 의 reviewer 위험 요소 모두 제거.

i Korean HLA n=874 DPB1*05:01 53.2% > Chu GD 44.0% · pooled OR 2.16 · 6/6 alleles direction- consistent ★ STRONG (격상 5/3)	ii GSE286332 mechanism 10,380 DEGs · 8-gene d=-1.60 · HLA-II d=+3.65 · IFN-γ FDR=2e-4 STRONG	iii Driver mRNA neutrality BRAF mRNA d=-0.04 (NS, V600E vs WT) · all driver AUC <0.61 · Driver_anchor ARI=0 STRONG	iv Pan-genome robustness TIERA67 ARI=0.90 ≈ pan-genome top- 5000 ARI=0.92 · 8- gene alone 0.49 (honest) STRONG	v Autoimmune-PTC HLA-II 140% mediation · TCGA Hashi-DM2 OR=0.20 p=6e-10 · TLS d=+1.96 · sub-B = NBNR STRONG
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THREE-LAYER DM1 MECHANISM (R5 V4 통합)

DM1 (cPTC-architected BRAF/RAS-neg) 은 (L1) **genetic** – 76.8% tyrosine kinase fusion (RET/NTRK/ALK/BRAF) + (L2) **mechanism heterogeneity** – fusion+ young/well-diff vs fusion- older/immune-

hot/Hashimoto-overlap + (L3) **epigenetic** – promoter HYPERMETHYLATION of differentiation machinery (TPO d=2.30, DIO1 d=1.24, TSHR d=1.20, fusion-independent).

II manuscript files · ~19,117 words total

First-pass drafts complete. Voice-protected sections (Hook/Aim/Disc 3.1/Limitations/Cover Para 1/Q9) = user 키보드.

#	File	Words	STATUS	VOICE
00	Title candidates 00_title_candidates.md	1,074	DRAFT V3	
01	Abstract (Cell Press structured) 01_abstract.md · 153 words target	804	V2	PROTECTED
02	Single-page outline (Mermaid + bullets) 02_outline.md · 5/4-6/13 schedule	4,805	V3	
03	Introduction (1.1-1.4) 03_introduction.md · 715 w target	1,368	V2	HOOK PROTECTED
04	Intro 1.1 Hook standalone 04_intro_1_1_hook.md	0	USER KEYBOARD	PROTECTED
04	Results (5 sub-results) 04_results.md · 3,250 w target	2,760	V3	
05	Figure 1-8 + Suppl S1-S9 captions 05_figure_captions.md	1,678	V2	
06	Discussion (3.1-3.4) + Limitations 06_discussion.md · 1,380 w target	1,756	V2	3.1 + LIMIT PROTECTED
07	STAR Methods (5 sub-sections) 07_star_methods.md	1,624	V2	
08	Cover letter + 5 suggested reviewers 08_cover_letter.md	909	V2	PARA 1 PROTECTED
09	12 Reviewer Q&A pre-empt 09_reviewer_qa.md · Q9 voice	1,401	V2	Q9 PROTECTED
11	Fig 7 (DM1 mechanism)	script	PYTHON	

	11_fig7_dm1_mechanism.py		
12	Fig 8 (Epigenetic methylation) 12_fig8_epigenetic.py	script	PYTHON
13	Supplementary tables S1-S9 13_supplementary_tables.md	938	V2

§ 3 · ABSTRACT KEY CLAIMS (153 WORDS TARGET)

4-section structured abstract

Cell Press 표준 – Background / Methods / Results / Conclusions.

Table 1. Abstract section breakdown (v2, 2026-04-30 — Conclusions (b) 채택, "dark matter" 따옴표 제거).

SECTION	WORDS V1	WORDS V2 *	CELL PRESS STANDARD
Background	20	22	20-30
Methods	42	40	35-40 (web Claude rec)
Results	60	66	60-70 (R5-2 보강)
Conclusions	25	25	20-30
Total	147	153	150 ± 5

COHORTS (COVER LETTER LEVEL)

TCGA-THCA n=504 + MSK-IMPACT n=117 + Korean cohorts n=874 + cBioPortal SV n=542 + HM450 methylation n=503. External sc validation: Pu 2021 (n=7) + Lu 2023 (n=23).

§ 4 · COVER LETTER – 5 SUGGESTED REVIEWERS

No conflicts of interest declared

Cover letter 의/ reviewer suggestion (Cell Rep Med standard 5명).

Table 2. Suggested reviewers from cover letter.

NAME	AFFILIATION	EXPERTISE
(content from 08_cover_letter.md – voice-protected)	Cover letter Paragraph 1 = user 키보드. 본 dashboard 는 navigation only.	

§ 5 · EXTERNAL NAVIGATION

Table 3. Paper 1 dashboard navigation.

RESOURCE	PATH	PURPOSE
Yu review dashboard (5-pillar visual)	../reports/yu_review_dashboard/	Phase 0 Cancer paper review dashboard, Pillar 별 figure embed
References.bib (17 cites)	03_intro_references.bib	Chu 2018 entry placeholder ⚠ user verify (REFERENCES_BIB_AUDIT)
Master papers index	../submission/papers_overview.html	6 papers 통합 dashboard
Paper 2 brief (autoimmune layer)	../manuscript_p2_brief/paper2_brief.html	Sister paper – DM2 autoimmune-overlap brief
R5 audit summary v9	../results/audit_2026_04_30/FINAL_COMPREHENSIVE_SUMMARY_v9.md	6-session audit (R1-R5) integration, 31 paper-shaping findings
Paper 1 README	README.md	Original directory navigation

§ 6 · 6-WEEK MARATHON (5/4-6/13)

Writing schedule + submission target 6/13

Pillar 1 STRONG = 분석 끝. 5/4-6/13 6주 manuscript writing marathon (메모리/v17_marathon_mode_post_pillar1).

Table 4. Paper 1 (Cancer v8) 6-week marathon schedule.

WEEK	PERIOD	FOCUS	DELIVERABLE
W1	5/4 – 5/10	Methods + Suppl freeze	STAR Methods (5 subsections) + Suppl tables S1-S9
W2	5/11 – 5/17	Results section completion	5 sub-results final, Fig 1-8 captions sync
W3	5/18 – 5/24	Discussion 3.1 voice (user)	3.1 mechanism + 3.2 clinical + 3.3 limitations + 3.4 future
W4	5/25 – 5/31	Cover letter + reviewer Q&A	Cover Para 1 voice + 12 Q&A pre-empt
W5	6/1 – 6/7	Figure final + reference verify	Fig 7 + Fig 8 Python, references.bib Chu 2018 verify
W6	6/8 – 6/13	Final compile + submit	Cell Rep Med submission package, bioRxiv preprint

Paper 1 (Cancer v8) – manuscript dashboard. Generated 2026-05-02. Source: Paper 1 README + manuscript file inventory.

Voice-protected sections (user 키보드): Hook (04_intro_1_1_hook) · Aim · Disc 3.1 (06_discussion) · Limitations · Cover Para 1 (08_cover_letter) · Q9 (09_reviewer_qa)

Master index: [../submission/papers_overview.html](#) (6 papers 통합)

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